

Supplementary Material

Hippocampal Segmentation and Measurement Protocols

Automated Segmentation

Hippocampal segmentation was performed using **FreeSurfer v7.1.1**. Specifically, we utilized the dedicated hippocampal/amygdala module (segmentHA_T1.sh), which employs a probabilistic atlas constructed from ultra-high resolution *ex vivo* MRI data (~0.1 mm isotropic) to segment the hippocampal substructures. The discrete voxel counts from the FreeSurfer output were aggregated into three levels of analysis: Local, Regional, and Composite.

1. Composite Protocols ("Whole" Level)

Composite measures were calculated to test hypothesis-driven anatomical groupings. These were derived by summing specific Local subfields defined in Table S1. Definitions follow the hierarchy proposed by *Roddy et al. (2019)*, alongside additional exploratory combinations.

Supplementary Table S1. Definition of Hippocampal Composite Measures.

Composite Name	Components (Sum of Local Subfields)	Description/Rationale
Global		
Hippocampal Extended (HE)	All 11 Subfields	The broadest definition; includes all segmented hippocampal tissue (excluding fissure).
Hippocampal Formation (HF)	HP + Dentate (GC-ML-DG) + Subiculum + Tail	Represents the functional hippocampal unit including input/output structures.
Hippocampal Proper (HP)	CA1 + CA3 + CA4	Restricted definition; includes only the Cornu Ammonis fields.
Exploratory		
Combined Dentate	CA4 + GC-ML-DG	Combines the polymorphic layer (CA4) and the granule cell layer (DG).
CA Only	CA1 + CA3	Exclusive Cornu Ammonis components without the dentate hilum (CA4).
Combined Dentate/CA	CA1 + CA3 + Combined Dentate	Represents the primary components of the trisynaptic circuit.

2. Local and Regional Protocols ("Parts" Level)

The **Local Protocol** (Transverse Axis) constructs histological subfields by summing the head and body components of a specific structure (e.g., CA1 = CA1-Head + CA1-Body). The **Regional Protocol** (Longitudinal Axis) constructs macro-regions by summing all substructures geographically located within the Head, Body, or Tail.

Supplementary Table 2. Definition of Local and Regional Protocols based on FreeSurfer labels.

Protocol	Measure	Calculation (Sum of FreeSurfer HBT Labels)
Local (Transverse)	Parasubiculum	Parasubiculum
	Presubiculum	Presubiculum-head + Presubiculum-body
	Subiculum	Subiculum-head + Subiculum-body
	CA1	CA1-head + CA1-body
	CA3 (incl. CA2)*	CA3-head + CA3-body
	CA4	CA4-head + CA4-body
	GC-ML-DG	GC-ML-DG-head + GC-ML-DG-body
	Molecular Layer	Molecular_layer_HP-head + Molecular_layer_HP-body
	HATA	HATA (Hippocampal-Amygdaloid Transition Area)
	Fimbria	Fimbria
	Tail	Hippocampal_tail
Regional (Longitudinal)	Head	Parasubiculum + HATA + (Head components of: Presubiculum, Subiculum, CA1, CA3, CA4, GC-ML-DG, Molecular_layer)
	Body	Fimbria + (Body components of: Presubiculum, Subiculum, CA1, CA3, CA4, GC-ML-DG, Molecular_layer)
	Tail	Hippocampal_tail

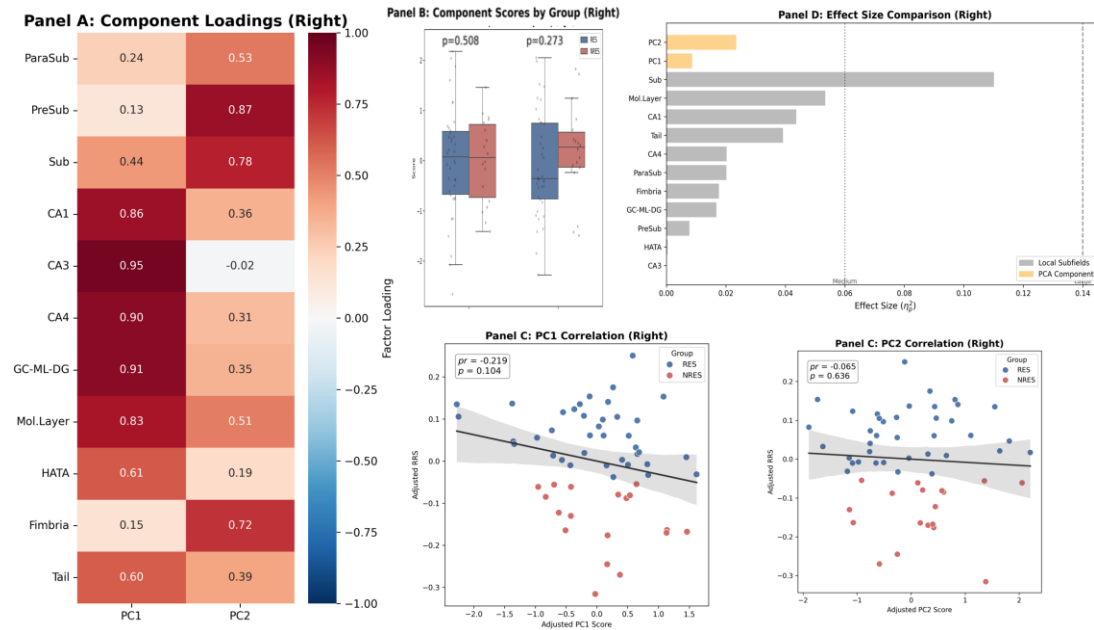
**Note: In FreeSurfer v7.1.1, the CA2 subfield is included within the CA3 label.*

**The Hippocampal Fissure was excluded from all volumetric calculations.*

Supplementary Table 3. Descriptive statistics for Hippocampal Volumes across all groups.

Region / Measure	Healthy Controls (N=70)	Responders (N=38)	Non-responders (N=18)
L-parasubiculum	69.50 (9.54)	68.24 (12.24)	66.79 (9.44)
L-presubiculum	308.66 (33.33)	302.29 (37.10)	307.76 (32.47)
L-subiculum	423.30 (45.67)	416.86 (43.13)	440.26 (44.48)
L-CA1	605.01 (60.84)	588.98 (70.28)	622.84 (56.57)
L-CA3	186.58 (25.63)	179.36 (25.49)	191.74 (21.17)
L-CA4	230.64 (19.72)	225.49 (23.74)	237.76 (19.45)
L-GC.ML.DG	269.08 (23.09)	261.56 (27.15)	274.53 (22.89)
L-molecular.layer	531.71 (44.07)	518.34 (54.08)	548.60 (44.51)
L-HATA	54.61 (7.41)	50.44 (7.18)	52.22 (6.76)
L-fimbria	85.32 (18.77)	84.92 (18.98)	87.12 (13.48)
L-tail	577.60 (67.66)	552.61 (64.88)	596.13 (56.29)
R-parasubiculum	66.36 (10.22)	65.89 (10.91)	62.17 (8.91)
R-presubiculum	292.14 (30.22)	291.39 (32.52)	294.81 (32.01)
R-subiculum	425.81 (44.37)	422.38 (40.71)	441.98 (39.02)
R-CA1	643.66 (75.12)	644.75 (77.22)	659.95 (62.37)
R-CA3	201.98 (28.11)	200.71 (28.73)	197.10 (24.31)
R-CA4	238.95 (23.36)	237.21 (23.49)	239.57 (20.17)
R-GC.ML.DG	277.74 (27.32)	275.64 (27.19)	277.67 (23.31)
R-molecular.layer	553.46 (50.93)	551.98 (54.30)	564.28 (43.94)
R-HATA	56.09 (8.96)	54.44 (7.93)	53.04 (7.50)
R-fimbria	82.26 (17.00)	82.84 (16.27)	86.67 (13.93)
R-tail	576.98 (69.74)	587.44 (65.24)	603.52 (46.96)
L-head	1624.55 (160.83)	1585.71 (182.89)	1654.49 (156.54)
L-body	1139.86 (90.89)	1110.77 (102.73)	1175.13 (84.63)
R-head	1685.96 (180.74)	1683.44 (175.77)	1712.67 (164.54)
R-body	1152.49 (103.67)	1143.80 (105.38)	1164.57 (81.48)
L-HE	3342.01 (269.25)	3249.09 (312.83)	3425.75 (259.01)
L-HF	2292.21 (192.75)	2224.87 (215.08)	2363.25 (182.26)
L-HP	1022.22 (97.28)	993.83 (111.78)	1052.33 (90.83)
L-Combined.Dentate	499.72 (42.65)	487.05 (50.73)	512.29 (42.18)
L-CA.Only	791.59 (80.27)	768.34 (90.96)	814.58 (74.01)
L-Combined.Dentate.CA	1291.31 (118.53)	1255.40 (136.93)	1326.86 (111.85)
L-CA2.4	417.21 (43.17)	404.85 (47.10)	429.50 (38.57)
R-HE	3415.43 (291.37)	3414.68 (315.68)	3480.76 (254.27)
R-HF	2365.11 (214.19)	2368.14 (228.55)	2419.80 (176.73)
R-HP	1084.59 (119.90)	1082.68 (123.26)	1096.63 (97.63)
R-Combined.Dentate	516.69 (50.47)	512.86 (50.50)	517.24 (43.19)
R-CA.Only	845.64 (98.50)	845.47 (101.58)	857.06 (80.32)
R-Combined.Dentate.CA	1362.33 (145.97)	1358.32 (149.38)	1374.30 (119.36)
R-CA2.4	440.93 (50.22)	437.93 (50.85)	436.67 (42.86)

Note: Values are presented as Mean (Standard Deviation) in mm³.



Supplementary Fig 1, Principle Component Analysis (PCA) of the local protocol on the right hemisphere. Two principal components (PC1 and PC2) were extracted. PC1 is heavily loaded by the core hippocampal subfields (CA, DG, Mol. Layer, HATA and Tail), while PC2 is loaded by the subiculum complex and fimbria(A). However, neither of these principal components differ between responders or non-responders (B), or show correlations with reduction rates of HAMD score (C). The effect size of PC1 and PC2 didn't surpass any of the subfield volume z-scores (D).

Supplementary Table 4. Partial correlation coefficients between hippocampal metrics and symptom improvement (HAMD RRS).

Measure	Left Hemisphere (pr)	Left P-value	Right Hemisphere (pr)	Right P-value	AI (pr)	AI P-value
Local Protocol						
ParaSub	-0.012	0.93	0.21	0.12	-0.246	0.067
PreSub	-0.207	0.126	-0.08	0.56	-0.154	0.258
Sub	-0.322	0.016	-0.287	0.032	-0.069	0.611
CA1	-0.36	0.006	-0.252	0.061	-0.184	0.174
CA3	-0.29	0.03	-0.191	0.159	-0.121	0.374
CA4	-0.273	0.042	-0.237	0.079	-0.078	0.566
GC-ML-DG	-0.268	0.046	-0.243	0.071	-0.077	0.573
Mol.Layer	-0.38	0.004	-0.296	0.027	-0.23	0.087
HATA	-0.214	0.112	0.042	0.76	-0.257	0.056
Fimbria	-0.009	0.947	-0.106	0.435	0.075	0.582
Tail	-0.318	0.017	-0.181	0.181	-0.223	0.098
Regional Protocol						
Head	-0.326	0.014	-0.24	0.075	-0.218	0.107
Body	-0.318	0.017	-0.25	0.063	-0.121	0.376
Tail	-0.318	0.017	-0.181	0.181	-0.223	0.098
Composites						
HE	-0.373	0.005	-0.264	0.049	-0.278	0.038
HF	-0.388	0.003	-0.284	0.034	-0.235	0.081
HP	-0.353	0.008	-0.256	0.057	-0.164	0.226
Comb.DG	-0.272	0.043	-0.241	0.073	-0.078	0.568
CA.Only	-0.361	0.006	-0.253	0.06	-0.179	0.186
DG+CA	-0.342	0.01	-0.257	0.056	-0.15	0.27
CA2-4	-0.297	0.026	-0.221	0.102	-0.103	0.451

Notes: Data represent partial correlation coefficients (pr) and p-values between the Reduction Rate of HAMD Scores (RRS) and hippocampal metrics (Z-scores for volumes; Z-scores for AI) controlling for age, sex and ICV.

Abbreviations: AI, Asymmetry Index; HF, Hippocampal Formation; HE, Hippocampal Extended; RRS, Reduction Rate of Scores.

Supplementary Table 5. Complete Predictive Performance Metrics for All Evaluated Feature Sets Using TabPFN Classifier

Feature Set	AUC	Sen	Spe	Acc	PPV	NPV
"Whole" Level (Composite Strategies)						
L-HF	0.718	0.553	0.944	0.679	0.955	0.500
AI-HF	0.675	0.500	0.944	0.643	0.950	0.472
AI-HE	0.651	0.658	0.722	0.679	0.833	0.500
L-Combined	0.645	0.579	0.722	0.625	0.815	0.448
Dentate/CA						
L-HP	0.637	0.605	0.667	0.625	0.793	0.444
L-HE	0.633	0.553	0.944	0.679	0.955	0.500
L-CA2-4	0.620	0.711	0.556	0.661	0.771	0.476
L-CA Only	0.617	0.368	0.889	0.536	0.875	0.400
L-Combined Dentate	0.608	0.316	0.944	0.518	0.923	0.395
AI-CA Only	0.575	0.395	0.889	0.554	0.882	0.410
AI-Combined	0.572	0.395	0.833	0.536	0.833	0.395
Dentate/CA						
AI-CA2-4	0.556	0.316	0.889	0.500	0.857	0.381
AI-HP	0.554	0.395	0.889	0.554	0.882	0.410
R-HE	0.510	0.474	0.944	0.625	0.947	0.459
AI-Combined Dentate	0.490	0.921	0.167	0.679	0.700	0.500
R-HF	0.477	0.474	0.944	0.625	0.947	0.459
R-CA Only	0.352	0.000	1.000	0.321	0.000	0.321
R-HP	0.291	0.105	0.944	0.375	0.800	0.333
R-Combined	0.291	0.105	0.944	0.375	0.800	0.333
Dentate/CA						
R-CA2-4	0.190	0.000	1.000	0.321	0.000	0.321
R-Combined Dentate	0.165	0.000	1.000	0.321	0.000	0.321
Regional (Hippocampal substructures based on the long-axis organization)						
Bilateral-body	0.703	0.658	0.778	0.696	0.862	0.519
AI-Body+ AI-Tail	0.702	0.789	0.667	0.750	0.833	0.600
L- tail	0.683	0.579	0.833	0.661	0.880	0.484
L-body	0.677	0.658	0.833	0.714	0.893	0.536
Bilateral- tail	0.665	0.632	0.833	0.696	0.889	0.517
L- Body+L-Tail)	0.640	0.474	0.944	0.625	0.947	0.459
Bilateral- head	0.538	0.447	0.889	0.589	0.895	0.432
L- head	0.522	0.474	0.889	0.607	0.900	0.444
R- body	0.465	0.289	0.889	0.482	0.846	0.372
R- tail	0.461	0.395	0.889	0.554	0.882	0.410
R- head	0.398	0.289	0.833	0.464	0.786	0.357
Local (Hippocampal substructures based on the transverse-axis organization)						
L-parasubiculum	0.113	0.000	1.000	0.321	0.000	0.321
L-presubiculum	0.238	0.184	0.833	0.393	0.700	0.326
L-subiculum	0.646	0.605	0.833	0.679	0.885	0.500

L-CA1	0.602	0.447	0.889	0.589	0.895	0.432
L-CA3	0.596	0.474	0.778	0.571	0.818	0.412
L-CA4	0.627	0.711	0.556	0.661	0.771	0.476
L-GC-ML-DG	0.595	0.421	0.778	0.536	0.800	0.389
L-molecular_layer	0.615	0.500	0.944	0.643	0.950	0.472
L-HATA	0.292	0.000	1.000	0.321	0.000	0.321
L-fimbria	0.120	0.974	0.056	0.679	0.685	0.500
L- tail	0.683	0.579	0.833	0.661	0.880	0.484
Top-3 Left Subfields	0.677	0.553	0.944	0.679	0.955	0.500
Top-5 Left Subfields	0.649	0.553	0.944	0.679	0.955	0.500
Latent-PC1 (Left)	0.617	0.526	0.944	0.661	0.952	0.486
Latent-PC1-2 (Left)	0.648	0.553	0.833	0.643	0.875	0.469

Note:

Feature sets are grouped by segmentation protocol and sorted by AUC within each category. Models were trained using the TabPFN classifier with a Leave-One-Out Cross-Validation (LOOCV) scheme.

AUC: Area Under the Receiver Operating Characteristic Curve.

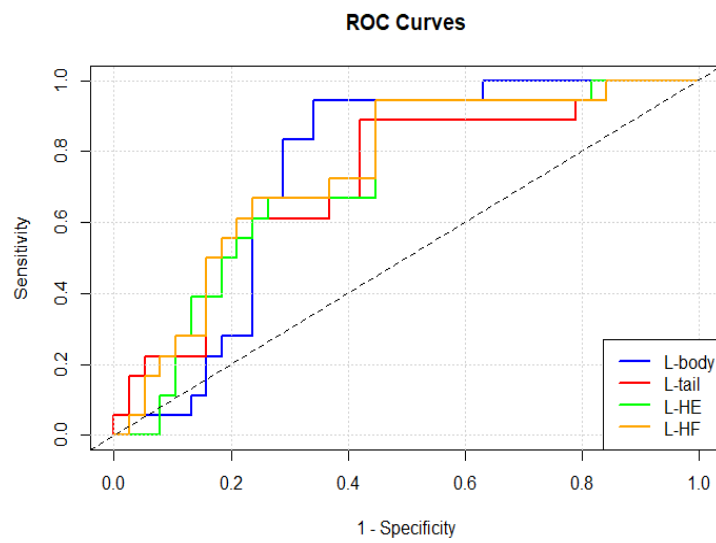
Sen./Spe./Acc.: Sensitivity, Specificity, and Accuracy calculated at the optimal decision threshold determined by Youden's Index.

PPV/NPV: Positive Predictive Value and Negative Predictive Value.

Top-3/Top-5 Left Subfields: Aggregated feature sets comprising the top 3 (Hippocampal Tail, Molecular Layer, Subiculum) and top 5 (plus CA4, CA1) left-sided local subfields, ranked by effect sizes in preliminary group comparisons.

Latent-PC: Principal Component scores derived from PCA on local subfields.

Abbreviations: AI, Asymmetry Index; CA, Cornu Ammonis; HE, Hippocampal Extended; HF, Hippocampal Formation; HP, Hippocampal Proper; L, Left; R, Right.



Supplementary Fig 2 ROC Curve Analysis with large effect size for differentiating responders from non-responders

Supplementary Table 6. ROC Curve Analysis for Differentiating the responders from nonresponders

	AUC	P	Sensitivity	Specificity	95% CI	cutoff
L-body	0.75	0.0008	65.79	94.44	0.616-0.856	≤ -21.012
L-tail	0.722	0.0039	57.89	88.89	0.586-0.833	≤ -8.5011
L-HE	0.73	0.0027	55.26	94.44	0.594-0.839	≤ -67.2467
L-HF	0.746	0.001	55.26	94.44	0.611-0.852	≤ -89.5581

Note: HE, Hippocampal Extended; HF, Hippocampal Formation